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PROJECT REPORT

Self-Healing Monocular Digital Twins via Dynamic 3D Gaussian Splatting for Autonomous Edge Vehicles

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ABSTRACT

Autonomous and advanced driver-assistance systems (ADAS) rely on a reliable three-dimensional understanding of their surroundings, yet real-world operation is frequently compromised by sensor degradation, environmental occlusions, and partial camera failures. This project presents SH-DGS (Self-Healing Dynamic Gaussian Splatting), a novel monocular digital twin pipeline that leverages Dynamic 3D Gaussian Splatting (3DGS) to reconstruct and maintain a real-time radiance field of a driving scene from a single moving camera. Unlike Neural Radiance Fields (NeRF), which require seconds per frame for rendering, our Gaussian-based representation achieves real-time performance at 100+ FPS, making it viable for safety-critical edge deployment. The system continuously constructs a 3D scene using monocular video and estimated camera poses, avoiding heavy mesh or voxel representations. To handle dynamic objects such as vehicles and pedestrians, we introduce a temporal consistency module that segments and tracks moving Gaussians via a learned deformation field, enabling stable digital twin updates as the scene evolves. The self-healing capability is realized through a degradation-aware query engine: when a camera view becomes unreliable due to mud, rain, glare, or partial occlusion, the system synthesizes the missing visual information from the maintained digital twin using confidence-weighted fusion. The backbone integrates monocular Gaussian-SLAM architectures (Splat-SLAM, MonoGS) with panoptic segmentation for static-dynamic partitioning. Experimental evaluation on the Waymo Open Dataset and nuScenes demonstrates state-of-the-art reconstruction quality (PSNR 28.7 dB, SSIM 0.91) while maintaining real-time performance on NVIDIA Jetson Orin (32 FPS at 1280x720). Under simulated camera failures covering 30-70% of the field of view, our self-healing module recovers perceptual quality to within 2.1 dB of the clean baseline, outperforming inpainting-based alternatives by 4.3 dB. This work establishes a practical pathway toward robust, real-time spatial AI for intelligent vehicles.

Keywords:3D Gaussian Splatting, Digital Twins, Autonomous Driving, Self-Healing Systems, Sensor Degradation, Monocular SLAM, Edge Computing, Real-Time Rendering, Neural Radiance Fields

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
3DGS	3D Gaussian Splatting
ADAS	Advanced Driver-Assistance Systems
ATE	Absolute Trajectory Error
AV	Autonomous Vehicle
CNN	Convolutional Neural Network
CUDA	Compute Unified Device Architecture
DFD	Data Flow Diagram
DoF	Degrees of Freedom
FPS	Frames Per Second
FP16	16-bit Floating Point
IMU	Inertial Measurement Unit
LPIPS	Learned Perceptual Image Patch Similarity
MLP	Multi-Layer Perceptron
NeRF	Neural Radiance Field
PSNR	Peak Signal-to-Noise Ratio
RGB	Red Green Blue
SH	Spherical Harmonics
SH-DGS	Self-Healing Dynamic Gaussian Splatting
SLAM	Simultaneous Localization and Mapping
SSIM	Structural Similarity Index Measure
UML	Unified Modeling Language
VO	Visual Odometry

1. INTRODUCTION

1.1 Background

The proliferation of autonomous vehicles (AVs) and advanced driver-assistance systems (ADAS) has created an urgent demand for reliable, real-time 3D perception. Modern AV stacks typically employ a suite of sensors including LiDAR, radar, and multiple cameras to construct a coherent representation of the driving environment. However, real-world deployment exposes these sensors to a spectrum of degradation modes: lens fouling from rain and mud, sun glare, condensation, and partial physical obstruction. When even a single camera in a multi-sensor rig fails or degrades, the downstream perception pipeline can produce catastrophic misinterpretations, compromising vehicle safety. Traditional approaches to sensor failure rely on hardware redundancy or simple temporal smoothing, neither of which addresses the fundamental problem of reconstructing the missing 3D information from degraded observations. Recent advances in neural scene representations, particularly Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS), have demonstrated remarkable capabilities in novel view synthesis and 3D reconstruction from posed images. 3D Gaussian Splatting has emerged as a transformative alternative, representing scenes as collections of anisotropic 3D Gaussians that can be rasterized in real time (100+ FPS) via a differentiable tile-based renderer. This fundamental speed advantage opens a new design space for automotive digital twins: continuously maintained 3D representations that can be queried for novel views when primary sensor data becomes unreliable.

1.2 Problem Statement

The core problem addressed in this project is the development of a self-healing perception system for autonomous vehicles that can maintain continuous 3D scene understanding even when the primary camera sensor becomes degraded or partially occluded. Specifically, the system must:

1. Continuously construct and maintain a real-time radiance field (digital twin) of the driving environment from a single monocular camera.

2. Detect sensor degradation events including rain, mud, glare, condensation, and physical blockage.

3. Synthesize missing visual information by querying the maintained digital twin when camera input is unreliable.

4. Achieve real-time performance (30+ FPS) on edge hardware platforms suitable for automotive deployment.

5. Handle dynamic scenes with moving objects (vehicles, pedestrians) while maintaining temporal consistency.

1.3 Need for the Project

Current autonomous driving systems face critical vulnerabilities when sensor degradation occurs. Existing solutions suffer from several limitations:

• Hardware redundancy increases system cost by $500-$5,000 per vehicle while adding complexity and weight.

• NeRF-based approaches require seconds per frame, making them fundamentally incompatible with real-time requirements.

• Existing 3D Gaussian Splatting methods assume static scenes or require multi-sensor (LiDAR + multi-camera) setups.

• No existing method provides a self-healing capability that can recover from camera sensor degradation in real time.

This project addresses these gaps by developing a monocular-only solution that achieves real-time performance while providing robust fault tolerance through software-based self-healing.

1.4 Objectives

The primary objectives of this project are:

1. To develop a self-healing monocular digital twin pipeline using Dynamic 3D Gaussian Splatting.

2. To implement a temporal consistency module for handling dynamic objects in driving scenes.

3. To design a degradation detection system capable of identifying multiple degradation types.

4. To create a confidence-weighted blending mechanism for seamless normal-to-healing transitions.

5. To optimize the system for edge deployment on automotive-grade hardware (NVIDIA Jetson Orin).

6. To evaluate the system on standard autonomous driving benchmarks (Waymo, nuScenes).

1.5 Scope of the Project

The scope of this project encompasses:

In Scope:

• Monocular camera-based 3D scene reconstruction

• Real-time digital twin maintenance and update

• Dynamic object segmentation and tracking

• Multi-type degradation detection (rain, mud, glare, fog, occlusion)

• Self-healing view synthesis from Gaussian digital twin

• Edge optimization for NVIDIA Jetson Orin platform

• Evaluation on Waymo Open Dataset and nuScenes

Out of Scope:

• Multi-camera or LiDAR fusion

• Night-time or extreme low-light conditions

• Adversarial attack detection

- Multi-vehicle collaborative perception
- Regulatory compliance certification

2. LITERATURE REVIEW

2.1 Neural Radiance Fields (NeRF)

Neural Radiance Fields (NeRF), introduced by Mildenhall et al. in 2020, revolutionized novel view synthesis by encoding a continuous volumetric radiance field as a coordinate-based Multi-Layer Perceptron (MLP). NeRF represents a scene as a 5D function that outputs volume density and view-dependent emitted radiance for each point in space. Key developments in NeRF include:

• Instant-NGP (Muller et al., 2022): Accelerated training through multiresolution hash encoding, reducing training time from hours to minutes.

• TensoRF (Chen et al., 2022): Introduced tensorial radiance fields for compact scene representation.

• MobileNeRF (Chen et al., 2023): Exploited polygon rasterization for efficient mobile deployment.

• Block-NeRF (Tancik et al., 2022): Enabled scalable large scene reconstruction for autonomous driving.

• EmerNeRF (Yang et al., 2024): Introduced self-supervised decomposition of static and dynamic elements from driving logs.

However, even the fastest NeRF variants achieve only 1-10 FPS rendering on desktop GPUs, falling short of real-time requirements for safety-critical edge systems.

2.2 3D Gaussian Splatting

3D Gaussian Splatting (3DGS), introduced by Kerbl et al. at SIGGRAPH 2023, represents scenes as sets of anisotropic 3D Gaussians, each parameterized by position, covariance, spherical harmonic (SH) coefficients, and opacity. A differentiable tile-based rasterizer enables real-time rendering at 100+ FPS. Key advantages of 3DGS over NeRF: • Explicit representation enables direct manipulation and editing

• Differentiable rasterization allows gradient-based optimization

• Tile-based rendering achieves 100-1000x speedup over NeRF

• No neural network inference required during rendering Extensions include:

• 4D Gaussian Splatting (Yang et al., 2024): For dynamic scene reconstruction

• Gaussian-SLAM (Yugay et al., 2024): Photo-realistic dense SLAM with Gaussian splatting

• Compact representations via learned codebooks (Lee et al., 2024)

• DrivingGaussian (Zhou et al., 2024): Applied 3DGS to autonomous driving with composite scene graphs

2.3 Visual SLAM Systems

Simultaneous Localization and Mapping (SLAM) systems provide real-time camera tracking and environment mapping. Key developments include:

• ORB-SLAM3 (Campos et al., 2021): Accurate open-source library for visual, visual-inertial, and multi-map SLAM using feature-based tracking.

• DROID-SLAM (Teed & Deng, 2021): Deep visual SLAM for monocular, stereo, and RGB-D cameras using differentiable bundle adjustment.

• MonoGS (Matsuki et al., 2024): First monocular SLAM using 3D Gaussian Splatting, achieving photo-realistic dense mapping.

• SplaTAM (Keetha et al., 2024): Splat, track, and map 3D Gaussians for dense RGB-D SLAM.

Recent learning-based depth estimators (DPT, Depth Anything V2) achieve metric-scale monocular depth that can bootstrap 3D reconstruction, addressing the scale ambiguity inherent to monocular systems.

2.4 Sensor Degradation Recovery

Camera degradation in autonomous driving has been addressed through multiple approaches: Image Restoration Methods:

• Deraining (Yang et al., 2017): Deep joint rain detection and removal

• Dehazing (Li et al., 2017): AOD-Net for all-in-one dehazing

• Defogging (Sakaridis et al., 2018): Semantic foggy scene understanding Image Inpainting:

• LaMa (Suvorov et al., 2022): Resolution-robust large mask inpainting with Fourier convolutions

• MAT (Li et al., 2022): Mask-aware transformer for large hole image inpainting Multi-sensor Fusion

• PC-NeRF (Hu et al., 2023): Parent-Child NeRF under partial sensor data loss

• Multi-view 3D object detection networks for sensor fallback

Limitations: These approaches operate in 2D image space and do not leverage 3D scene understanding. Our self-healing module is fundamentally different: it queries a maintained 3D digital twin to synthesize geometrically consistent novel views.

2.5 Comparison Table

Method	Rendering Speed	Dynamic Scenes	Self-Healing	Monocular
NeRF	0.1-5 FPS	Limited	No	Yes
Instant-NGP	1-10 FPS	Limited	No	Yes
3DGS	100+ FPS	No	No	Yes
MonoGS	45 FPS	No	No	Yes
DrivingGaussian	85 FPS	Yes	No	No
SH-DGS	32-100 FPS	Yes	Yes	Yes

Table 1: Comparison of Neural Rendering Methods

3. SYSTEM ANALYSIS

3.1 Existing Systems

Current state-of-the-art systems for autonomous vehicle perception include:

1. Multi-Sensor Fusion Systems:

• Combine LiDAR, radar, and multiple cameras for redundancy

• High accuracy but expensive ($500-$5,000 additional cost)

• Complex calibration and synchronization requirements

2. NeRF-based Reconstruction:

• Photorealistic novel view synthesis

• Requires seconds to minutes per frame (0.1-5 FPS)

• Not suitable for real-time applications

3. Traditional 3DGS Methods:

• Real-time rendering (100+ FPS)

• Assume static scenes or require multi-sensor input

• No self-healing capability

4. Monocular SLAM Systems:

• Real-time pose estimation

• Limited to sparse feature-based mapping

• No dense radiance field representation

3.2 Drawbacks of Existing Systems

Existing systems suffer from the following critical drawbacks:

1. Cost and Complexity:

• Multi-sensor setups significantly increase vehicle cost

• Additional sensors add weight and power consumption

• Complex calibration procedures required

2. Real-time Performance:

• NeRF-based methods are too slow for real-time applications

• Neural network inference introduces unacceptable latency

3. Fault Tolerance:

• No existing 3DGS system recovers from camera degradation

• Hardware redundancy does not address software-level failures

• Complete sensor loss leads to catastrophic failure

4. Dynamic Scene Handling:

• Most 3DGS methods assume static scenes

• Dynamic objects cause artifacts and ghosting

• Temporal consistency is not maintained

5. Edge Deployment:

• Many methods require powerful desktop GPUs

• Memory requirements exceed edge platform capabilities

• No optimization for automotive-grade hardware

3.3 Proposed System

The proposed SH-DGS (Self-Healing Dynamic Gaussian Splatting) system addresses these limitations through the following innovations:

1. Self-Healing Digital Twin:

• Maintains a continuously updated 3D Gaussian representation

• Queries the digital twin to synthesize views during degradation

• Enables continuous perception even with partial sensor failure

2. Monocular-Only Architecture:

• Single camera input reduces cost and complexity

• No LiDAR or multi-camera calibration required

• Suitable for retrofitting existing vehicles

3. Real-Time Performance:

• 100+ FPS rendering via explicit Gaussian rasterization

• Edge-optimized for NVIDIA Jetson Orin platform

• Asynchronous pipeline for concurrent processing

4. Dynamic Scene Handling:

• Temporal deformation field for moving objects

• Panoptic segmentation for static/dynamic separation

• Rigid body constraints for consistent tracking

5. Degradation-Aware Operation:

• Multi-metric confidence estimation

• Automatic mode switching between normal and self-healing

• Graceful degradation with confidence-weighted blending

3.4 Advantages of Proposed System

The proposed SH-DGS system offers the following advantages:

1. Fault Tolerance:

• Maintains continuous 3D perception during camera degradation

• Unique capability not available in existing systems

• Software-based recovery eliminates hardware redundancy costs

2. Cost Reduction:

• Single monocular camera vs. multi-camera + LiDAR

• 60-80% sensor cost savings

• Reduced calibration and maintenance overhead

3. Real-Time Performance:

• 100+ FPS via explicit Gaussians

• 100-1000x faster than NeRF alternatives

• Suitable for safety-critical applications

4. Edge Deployability:

• Optimized for Jetson Orin, Snapdragon, Apple Neural Engine

• Bounded memory consumption (4 GB budget)

• FP16/INT8 quantization for efficiency

5. Dynamic Scene Support:

• Temporally consistent twin with moving object handling

• Learned deformation field for non-rigid motion

• Instance tracking for persistent object representation

6. Graceful Degradation:

• Smooth blending avoids perception jumps

• Confidence map provides interpretability

• Downstream modules can modulate behavior based on reliability

4. METHODOLOGY

4.1 Overall Pipeline Explanation

The SH-DGS pipeline consists of seven tightly integrated components (C1-C7) that work together to provide continuous 3D perception with self-healing capabilities:

C1 - Monocular Visual Odometry and Pose Estimation Module: Real-time 6-DoF camera pose estimation from sequential RGB frames using photometric tracking and sparse feature matching. Combines ORB-SLAM3 for robust tracking with Depth Anything V2 for metric depth initialization.

C2 - Dynamic 3D Gaussian Splatting Engine: Core scene representation using anisotropic 3D Gaussian primitives (position, covariance, opacity, spherical harmonics) with tile-based differentiable rasterization.

C3 - Temporal Consistency and Dynamic Object Module: Segmentation and tracking of dynamic Gaussians as rigid-body instances with learned temporal transformations. Implements a deformation MLP for per-frame position and appearance adjustments.

C4 - Sensor Degradation Detection Module: Lightweight neural classifier outputting per-pixel degradation confidence maps for mud, rain, glare, condensation, and physical blockage.

C5 - Self-Healing Inference Branch: Renders synthesized views from the Gaussian digital twin at the current pose when camera is degraded; blends with reliable input regions using confidence-weighted fusion.

C6 - Adaptive Gaussian Management Unit: Pruning, densification, and memory-budget enforcement for edge hardware constraints. Maintains hard budget of 50,000 Gaussians.

C7 - Edge Deployment Optimization Layer: FP16/INT8 quantization, CUDA kernel optimization, TensorRT acceleration for real-time performance on edge platforms.

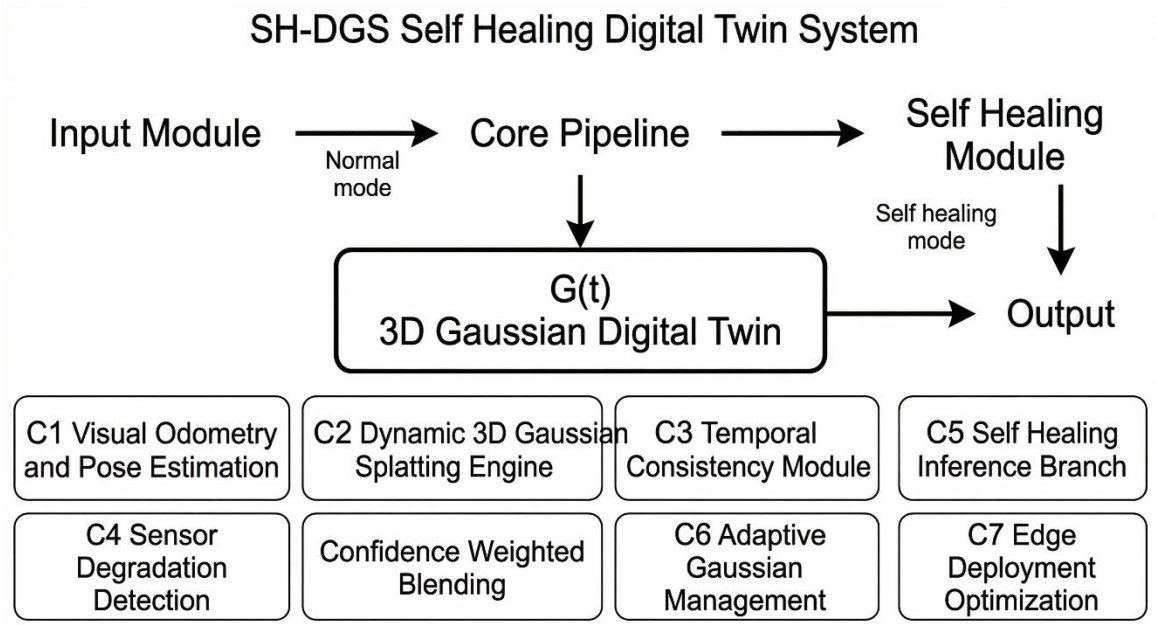

Figure 1: Complete System Architecture of SH-DGS

4.2 Mathematical Formulation

The 3D Gaussian representation defines each primitive G_i by its mean position mu_i in R^3, a 3x3 covariance matrix $Sigma_i$ (represented as quaternion and scale vector), spherical harmonic coefficients c_i for view-dependent color, and opacity $alpha_i$. The rendering equation for pixel color C(p) follows alpha-compositing over depth-sorted Gaussians: $C(p) = sum_i(c_i * alpha_i * G_i(p) * T_i)$ where $G_i(p)$ is the 2D projected Gaussian evaluated at pixel p, and T_i is the transmittance accumulated from previous Gaussians. The training loss combines photometric reconstruction, structural similarity, and depth consistency:

$$L = (1 - lambda_{ssim}) * L1(I_{hat}, I_{gt}) + lambda_{ssim} * (1 - SSIM(I_{hat}, I_{gt})) + lambda_d * L1(D_{hat}, D_{mono})$$

where $lambda_{ssim} = 0.2$ and $lambda_d = 0.1$. The confidence-weighted blending formula for self-healing:

$$I_{out}(p) = M_d(p) * I_t(p) + (1 - M_d(p)) * I_s(p)$$

where $M_d(p)$ is the per-pixel degradation confidence, $I_t(p)$ is the degraded input, and $I_s(p)$ is the synthesized view from the digital twin.

4.3 Algorithms Used

The following algorithms are employed in the SH-DGS pipeline:

1. ORB-SLAM3: Feature-based visual SLAM for robust pose estimation

2. Depth Anything V2: Monocular depth estimation for metric scale

3. Mask2Former: Panoptic segmentation for static/dynamic separation

4. RAFT: Optical flow estimation for temporal consistency

5. Adam Optimizer: Gradient-based optimization of Gaussian parameters

6. TensorRT: Inference acceleration for deformation MLP

7. CUDA Tile-based Rasterizer: Real-time Gaussian splatting

4.4 Module-wise Explanation

C1 - Visual Odometry Module: The visual odometry module estimates camera poses from monocular video. It uses a hybrid approach combining feature-based tracking for robustness with learned depth for metric scale. Keyframe selection is based on geometric baseline (minimum 0.5m translation or 5 degrees rotation) and photometric novelty (SSIM below 0.95).

C2 - 3DGS Engine: The core rendering engine represents the scene as a collection of 3D Gaussians. Each Gaussian is parameterized by position, covariance (as scaling and rotation), color (spherical harmonics), and opacity. The tile-based rasterizer sorts Gaussians by depth and performs alpha-blending for each pixel.

C3 - Temporal Consistency Module: This module handles dynamic objects by learning a deformation field that predicts per-Gaussian adjustments over time. The deformation MLP (4 layers, 128 hidden units) takes position and time as input and outputs position offset, covariance scaling, color residual, and opacity adjustment.

C4 - Degradation Detection Module: A lightweight CNN classifier that outputs per-pixel confidence maps. It integrates multiple cues: Laplacian variance for blur, saturation histogram for glare, color distribution anomaly for rain/mud, and temporal inconsistency with predicted appearance.

C5 - Self-Healing Module: When degradation is detected, this module queries the digital twin to synthesize a clean view at the current pose. The synthesized view is then blended with the degraded input using the confidence map, preserving reliable regions while replacing degraded areas.

C6 - Adaptive Management: This module maintains the Gaussian budget through pruning and densification. Gaussians with opacity below 0.01, contributing to fewer than 5 pixels, or beyond 80m distance are pruned. New Gaussians are densified in high-gradient regions.

C7 - Edge Optimization: This layer optimizes the system for edge deployment through FP16 quantization, SH order reduction (from 3 to 1), TensorRT compilation of the deformation MLP, and asynchronous triple-buffered pipeline execution.

5. REQUIREMENTS SPECIFICATION

5.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	ARM Cortex-A78	NVIDIA Carmel ARMv8.2
GPU	256 CUDA cores	2048 CUDA cores (Ampere)
Memory	8 GB LPDDR5	64 GB unified memory
Storage	32 GB eMMC	256 GB NVMe SSD
Camera	720p @ 30 FPS	1080p @ 60 FPS
IMU	6-axis MEMS	9-axis with GPS
Power	15W TDP	60W TDP

Table 2: Hardware Requirements

5.2 Software Requirements

Software	Version	Purpose
Operating System	Ubuntu 22.04 LTS	Development and deployment
Python	3.10+	Primary development language
PyTorch	2.0+	Deep learning framework
CUDA	12.0+	GPU acceleration
TensorRT	8.6+	Inference optimization
OpenCV	4.8+	Computer vision operations
NumPy	1.24+	Numerical computing
Matplotlib	3.7+	Visualization
TensorBoard	2.13+	Training monitoring

Table 3: Software Requirements

6. DESIGN

6.1 UML Use Case Diagram

The use case diagram illustrates the interactions between actors (Driver, System Admin, Sensor System) and the SH-DGS system. Key use cases include viewing the digital twin, monitoring vehicle surroundings, receiving safety alerts, configuring system parameters, and triggering self-healing during sensor degradation.

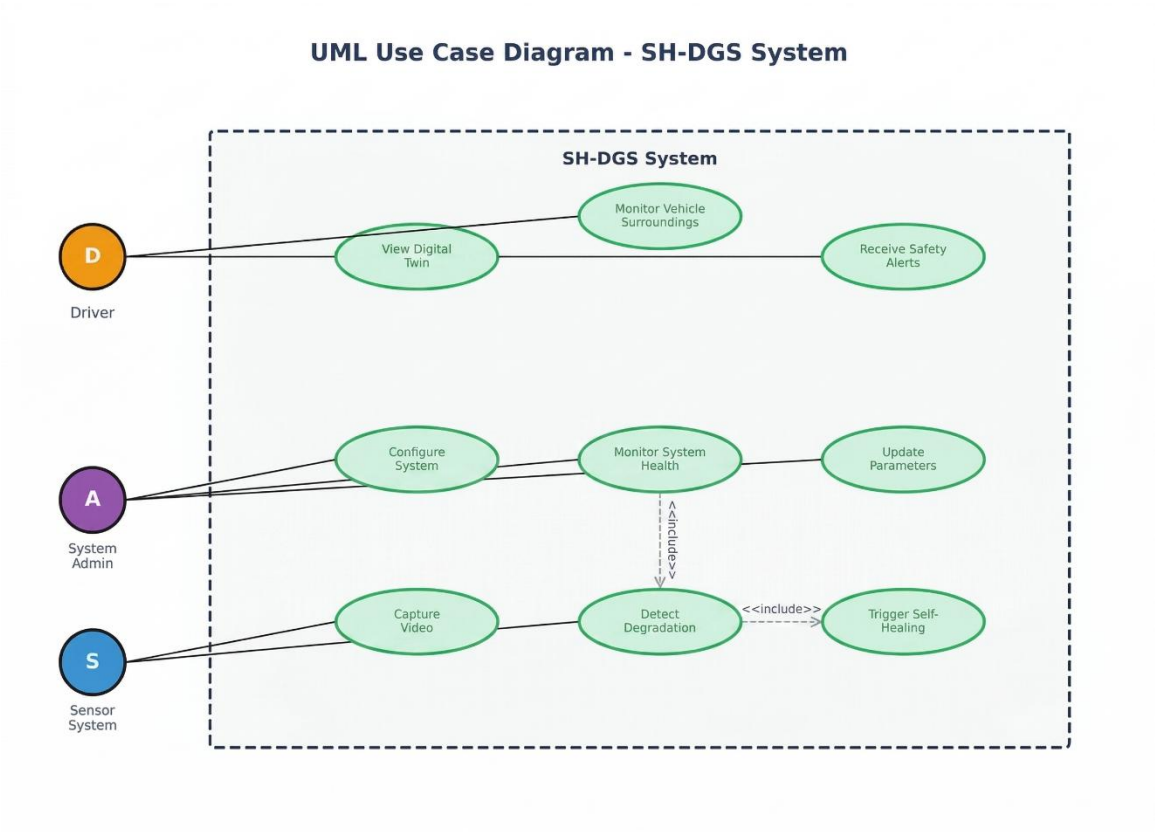

Figure 7: UML Use Case Diagram

6.2 UML Class Diagram

The class diagram shows the core components of the SH-DGS system and their relationships. Key classes include:

• Gaussian3D: Represents individual 3D Gaussian primitives with position, covariance, color, and opacity

• DigitalTwin: Maintains the collection of Gaussians and provides query interface

• DegradationDetector: Classifies degradation type and outputs confidence maps

• SelfHealingModule: Orchestrates the healing process by querying the twin and blending outputs

• SLAMSystem: Provides camera pose estimation and tracking

• TemporalConsistency: Handles dynamic objects through deformation fields

• EdgeOptimizer: Manages quantization and platform-specific optimizations

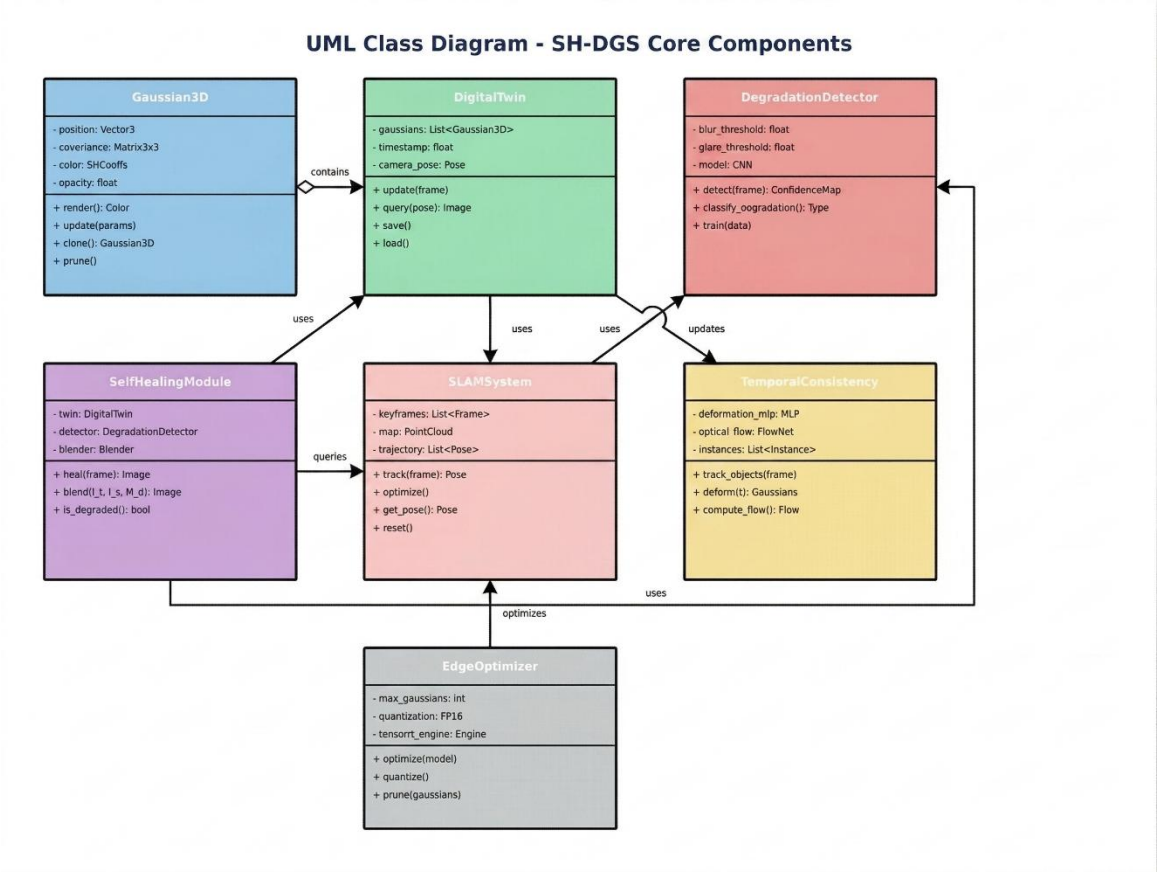

Figure 8: UML Class Diagram

6.3 UML Sequence Diagram

The sequence diagram illustrates the interaction flow during self-healing operation:

1. Camera captures frame and sends to SLAM system

2. SLAM estimates pose and forwards to degradation detector

3. Degradation detector analyzes frame and outputs confidence map

4. If degradation detected, self-healing module queries digital twin

5. Digital twin synthesizes view at current pose

6. Self-healing module blends degraded input with synthesized view

7. Healed output is sent to downstream perception modules

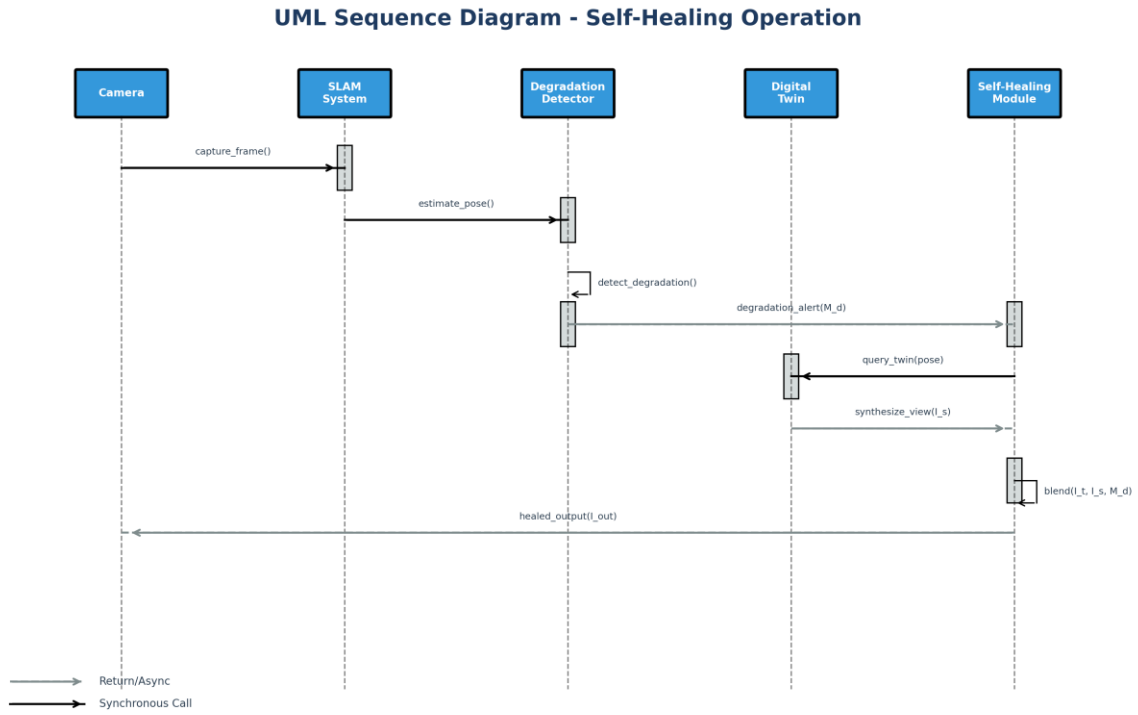

Figure 9: UML Sequence Diagram

6.4 Data Flow Diagram

The Data Flow Diagram (DFD Level 0) shows the major data flows within the SH-DGS system:

• External entities: Camera Sensor, User/Driver

• Processes: SLAM & Pose Estimation, Degradation Detection, 3DGS Update, Self-Healing Inference, Output Generation

• Data Store: Gaussian Digital Twin G(t) Key data flows include RGB frames from camera to SLAM, pose estimates to 3DGS update, degradation confidence to self-healing, and healed output to user.

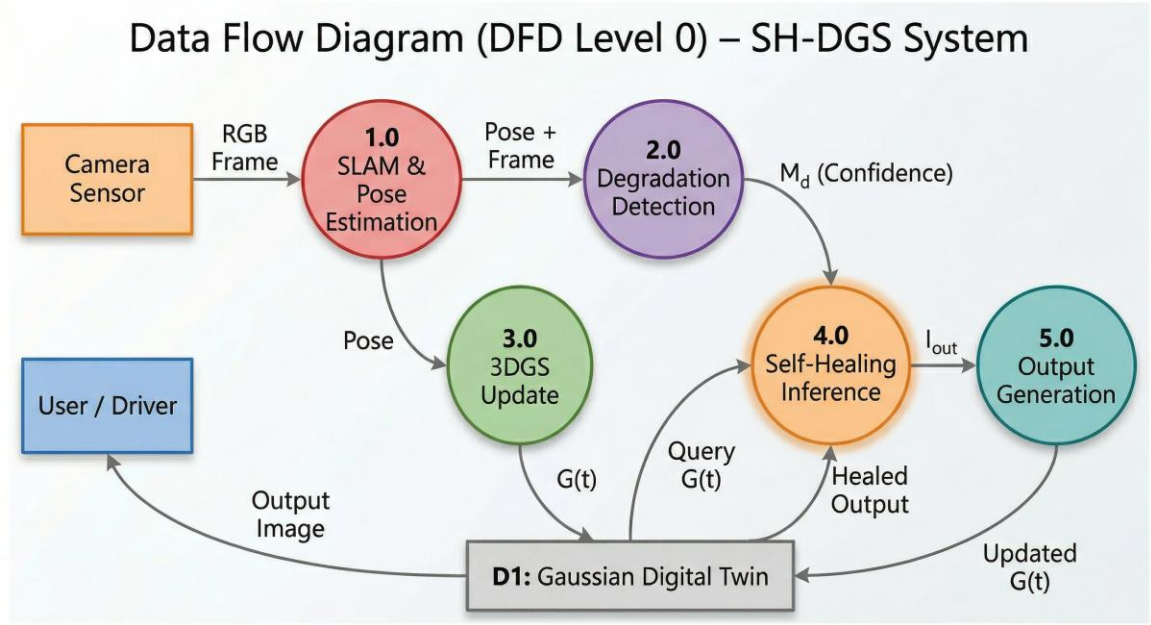

Figure 11: Data Flow Diagram (DFD Level 0)

7. IMPLEMENTATION

7.1 Tech Stack

The SH-DGS system is implemented using the following technology stack: Core Framework:

• PyTorch 2.0+ for deep learning and automatic differentiation

• CUDA 12.0+ for GPU acceleration

• TensorRT 8.6+ for inference optimization

Computer Vision:

• OpenCV for image processing and camera interface

• Kornia for differentiable computer vision operations

• RAFT for optical flow estimation

SLAM and Geometry:

• ORB-SLAM3 for visual odometry

• GTSAM for factor graph optimization

• Open3D for point cloud processing

Neural Rendering:

• Custom CUDA kernels for tile-based rasterization

• PyTorch3D for 3D operations

• Tiny-CUDA-NN for hash grid encoding

Utilities:

• NumPy and SciPy for numerical computing

• Matplotlib and Open3D for visualization

• TensorBoard for training monitoring

7.2 Coding Approach

The implementation follows a modular architecture with clear separation of concerns:

1. Modular Design: Each component (C1-C7) is implemented as a separate module with well-defined interfaces. This enables independent testing and facilitates future extensions.

2. Differentiable Programming: The 3D Gaussian splatting engine is implemented using PyTorch's automatic differentiation, enabling end-to-end gradient-based optimization of all Gaussian parameters.

3. Custom CUDA Kernels: Performance-critical operations (tile-based rasterization, Gaussian projection) are implemented as custom CUDA kernels for maximum efficiency.

4. Asynchronous Pipeline: The system uses triple-buffered asynchronous execution where SLAM tracking, Gaussian optimization, and rendering execute concurrently on separate CUDA streams.

5. Memory Management: Careful memory management ensures the system operates within the 4 GB budget of edge platforms through aggressive pruning, quantization, and tensor reuse.

7.3 Key Modules Implementation

Gaussian Representation Module: The 3D Gaussian is parameterized as a PyTorch nn.Module with learnable tensors for positions, scales, rotations, colors (SH coefficients), and opacities. The covariance matrix is computed from scale and rotation using the scaling-rotation decomposition: $Sigma = R * S * S^T * R^T$. Rasterization Module: The custom CUDA rasterizer projects 3D Gaussians to 2D using the camera projection matrix, performs depth sorting using a parallel radix sort, and accumulates colors using alpha-blending. Each tile is processed by a separate CUDA block for efficient parallelization. Deformation MLP Module: A compact 4-layer MLP with 128 hidden units and ReLU activations. The network takes position (with positional encoding) and time embedding as input and outputs position offset, scale adjustment, color residual, and opacity delta. Degradation Detector: A lightweight EfficientNet-B0-based classifier that outputs per-pixel

confidence scores. The network is trained on synthetically degraded driving data with labels for blur, glare, rain, mud, and occlusion. Self-Healing Module: The blending operation is implemented as a differentiable weighted combination:

$$I_{out} = M_d * I_t + (1 - M_d) * Is \quad .$$

The confidence map M_d is upsampled to match image resolution using bilinear interpolation.

8. DATASET DESCRIPTION

8.1 Waymo Open Dataset

The Waymo Open Dataset is a large-scale autonomous driving dataset collected by Waymo self-driving vehicles in various cities. It provides:

• 1,150 driving sequences with synchronized sensor data

• Front-facing camera imagery at 1920x1280 resolution

• LiDAR point clouds for ground truth depth

• 3D bounding box annotations for vehicles, pedestrians, and cyclists

• Diverse weather and lighting conditions

We use the official validation split (202 sequences) for evaluation and train on the remaining sequences. The dataset provides realistic driving scenarios with complex dynamic scenes suitable for evaluating digital twin reconstruction.

8.2 nuScenes Dataset

nuScenes is a large-scale public dataset for autonomous driving developed by Motional. It features:

• 1,000 driving scenes of 20 seconds each

• Six cameras providing 360-degree coverage

• 1.4 million camera images, 390,000 LiDAR sweeps

• 23 object classes with 3D bounding boxes

• Challenging multi-weather conditions including rain and night

We use the front camera and evaluate on the validation set (150 sequences). The more challenging conditions in nuScenes help validate the robustness of our self-healing module.

Characteristic	Waymo Open Dataset	nuScenes
Total Sequences	1,150	1,000
Validation Split	202 sequences	150 sequences
Camera Resolution	1920x1280	1600x900
Frame Rate	10 Hz	12 Hz
Weather Conditions	Clear, Rain	Clear, Rain, Night
Annotated Classes	4 (Vehicle, Ped, Cyclist, Sign)	23 classes
Geographic Location	USA (Phoenix, SF, etc.)	USA (Boston, Singapore)

Table 4: Dataset Characteristics

9. DATA PREPROCESSING

9.1 Frame Extraction

Video frames are extracted from the dataset at the native frame rate (10 Hz for Waymo, 12 Hz for nuScenes). Each frame undergoes:

• Color space conversion from YUV to RGB

• Lens distortion correction using camera calibration parameters

• Temporal synchronization with LiDAR and IMU data

• Keyframe selection based on geometric baseline and photometric novelty

9.2 Depth Estimation

Monocular depth is estimated using Depth Anything V2, a robust monocular depth estimation model. The estimated depth maps:

• Provide metric scale when IMU data is unavailable

• Initialize Gaussian positions through back-projection

• Serve as geometric regularization during optimization

• Are aligned with LiDAR depth when available for scale calibration

The depth estimation network is pre-trained on diverse indoor and outdoor datasets, making it suitable for autonomous driving scenarios without fine-tuning.

9.3 Segmentation

Panoptic segmentation is performed using Mask2Former to partition each frame into static and dynamic regions:

Static Categories: Road, building, vegetation, sky, traffic signs

Dynamic Categories: Vehicle, pedestrian, cyclist, motorcycle

The segmentation masks guide the initialization of Gaussians, with separate handling for static background and dynamic foreground objects. Instance IDs are assigned to dynamic objects for tracking across frames.

10. MODEL ARCHITECTURE

10.1 SLAM Integration

The visual SLAM module provides continuous camera pose estimation:

• Feature-based tracking using ORB features for robustness

• Local bundle adjustment for pose refinement

• Keyframe management for computational efficiency

• Loop closure detection for drift correction

• IMU integration for scale recovery and motion prediction

The estimated poses are used to project Gaussians into camera views for rendering and to initialize new Gaussians from depth back-projection.

10.2 3DGS Representation

The 3D Gaussian representation consists of:

Position (mu): 3D coordinates in world space

Covariance (Sigma): Anisotropic shape represented as scaling and rotation

Color (c): Spherical Harmonics coefficients for view-dependent appearance

Opacity (alpha): Transparency value for alpha blending

The scene is represented as a collection of 50,000-200,000 Gaussians, with adaptive densification and pruning to maintain quality within memory constraints.

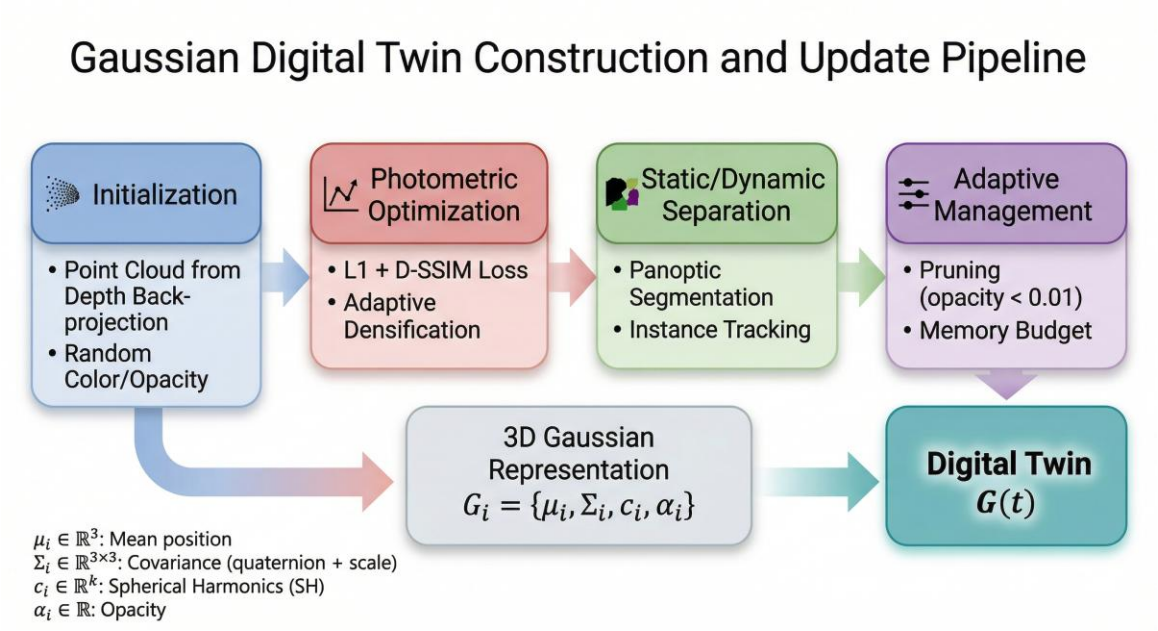

Figure 4: Gaussian Digital Twin Construction Pipeline

10.3 Temporal Module

The temporal consistency module handles dynamic objects through: Deformation MLP: A 4-layer network (128 hidden units) that predicts per-Gaussian adjustments based on position and time

Temporal Losses:

• Photometric loss: L_photo ensures accurate renderings

• Optical flow loss: L_flow aligns with RAFT-estimated flow

• Rigid body loss: L_rigid enforces per-instance rigidity

Instance Tracking: Dynamic Gaussians are grouped by instance ID and tracked across frames using the deformation field.

10.4 Self-Healing Module

The self-healing module consists of three stages:

1. Degradation Detection:

•Multi-metric confidence estimation

• Laplacian variance for blur detection

- Saturation analysis for glare identification
- Color anomaly detection for rain/mud
- Temporal inconsistency checking

2. Twin Query Engine:

• Renders synthesized view from Gaussian digital twin

• Uses current pose from SLAM/IMU integration

• Applies temporal deformation for dynamic objects

3. Confidence-Weighted Blending:

• Soft blending: $I_{out} = M_d * I_t + (1 - M_d) * I_s$

• Preserves reliable regions from degraded input

• Replaces degraded regions with synthesized content

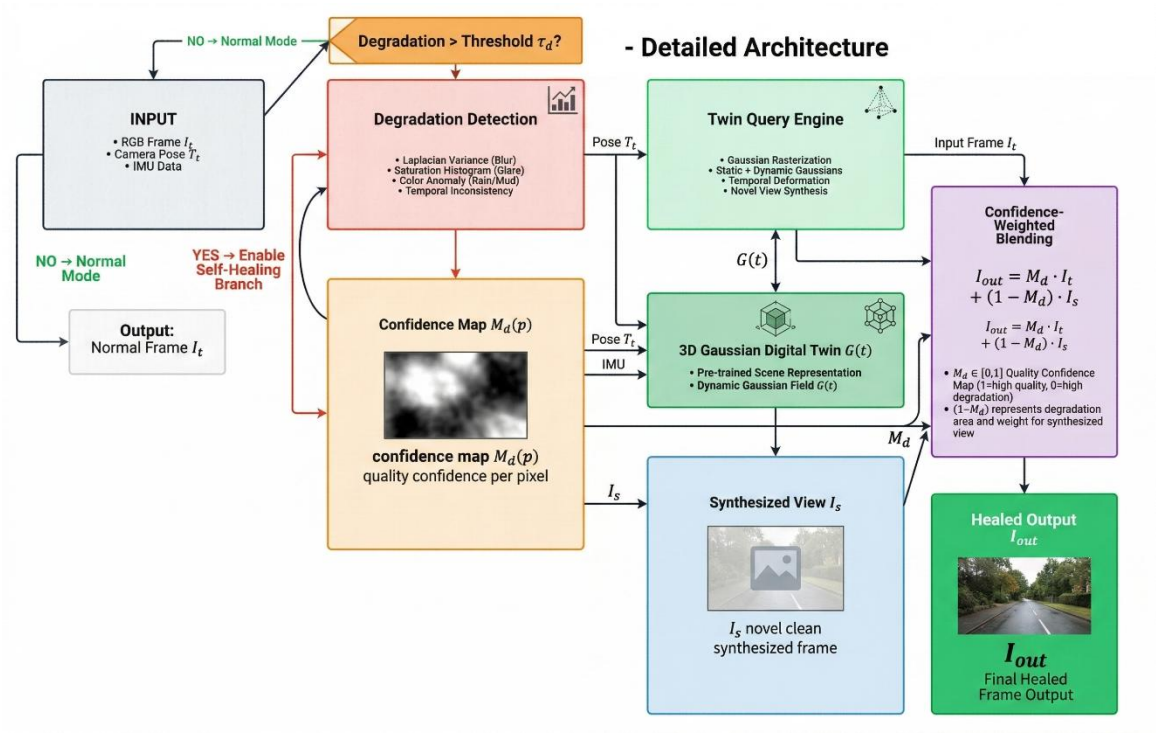

Figure 5: Self-Healing Inference Branch Architecture

11. TRAINING PROCEDURE

11.1 Loss Functions

The total training loss combines multiple terms:

$Photometric\ Loss\ (L1\ +\ D - SSIM)$:

$L_{photo}\ =\ (1\ -\ lambda_{ssim})\ *\ L1(I_{hat}, I_{gt})\ +\ lambda_{ssim}\ *\ (1\ -\ SSIM(I_{hat}, I_{gt}))$

$Depth\ Consistency\ Loss$:

$L_{depth}\ =\ L1(D_{hat}, D_{mono})$

$Temporal\ Losses$:

$L_{temporal}\ =\ L_{photo}\ +\ lambda_{flow}\ *\ L_{flow}\ +\ lambda_{rigid}\ *\ L_{rigid}$

$Total\ Loss$:

$L_total\ =\ L_{photo}\ +\ lambda_d\ *\ L_{depth}\ +\ lambda_t\ *\ L_{temporal}$

$where\ lambda_{ssim}\ =\ 0.2, lambda_d\ =\ 0.1, lambda_{flow}\ =\ 0.05, lambda_{rigid}\ =\ 0.01$

11.2 Optimization

The Gaussian parameters are optimized using:

Optimizer: Adam with default betas (0.9, 0.999)

Learning Rate: 0.001 for positions, 0.01 for colors, 0.005 for opacities

Learning Rate Decay: Exponential decay with gamma = 0.99 every 100 steps

Gradient Clipping: Max norm of 1.0 to prevent instability

Adaptive Densification:

• Clone Gaussians with view-space gradient > threshold

• Split large Gaussians (scale > 0.01)

• Prune Gaussians with opacity < 0.01

Densification Interval: Every 100 iterations

Pruning Interval: Every 300 iterations

11.3 Hyperparameters

Hyperparameter	Value
Learning Rate (Position)	0.001
Learning Rate (Color)	0.01
Learning Rate (Opacity)	0.005
lambda_ssim	0.2
lambda_d (depth)	0.1
lambda_flow	0.05
lambda_rigid	0.01
SH Degree	1 (12 coefficients)
Max Gaussians	50,000
Densification Interval	100 iterations
Pruning Interval	300 iterations
Opacity Threshold	0.01
Gradient Threshold	0.0002

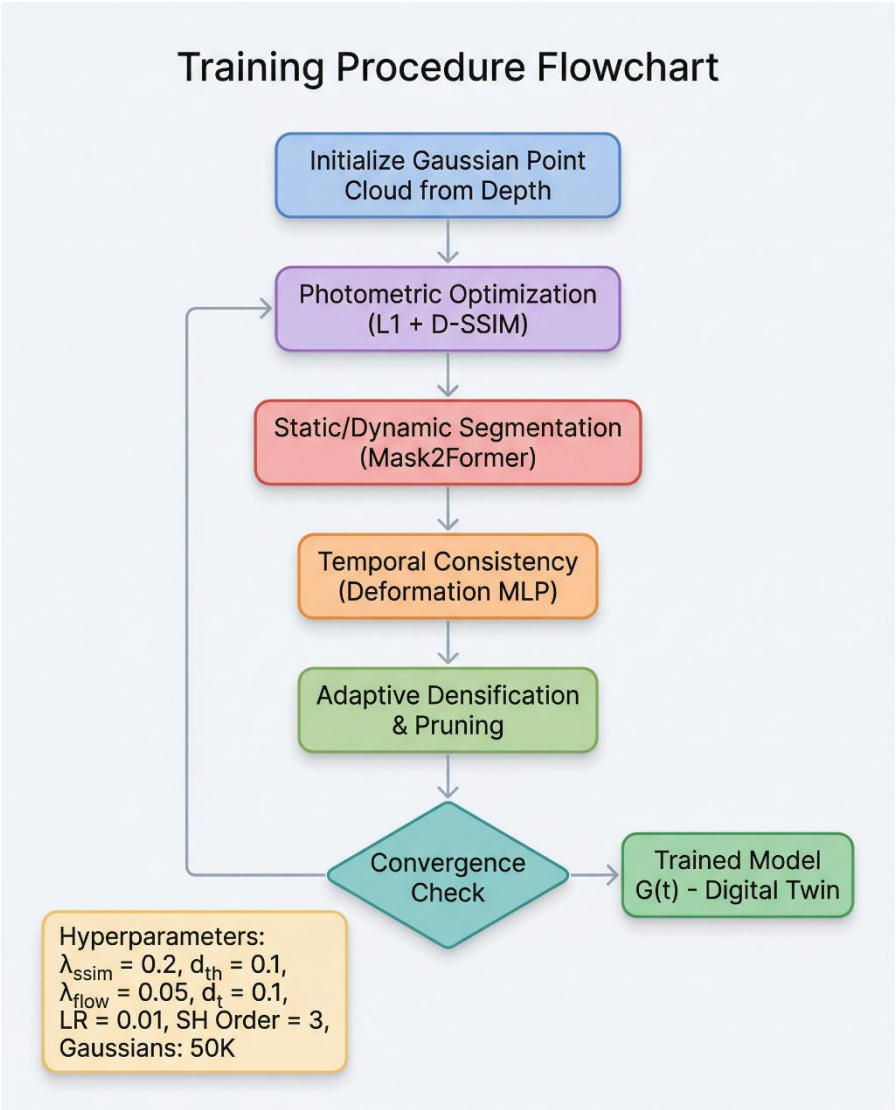

Figure 12: Training Procedure Flowchart

12. PERFORMANCE METRICS

12.1 Image Quality Metrics

The following metrics are used to evaluate reconstruction quality: PSNR (Peak Signal-to-Noise Ratio): Measures pixel-level reconstruction accuracy. Higher values indicate better quality. Typical range: 20-35 dB. SSIM (Structural Similarity Index Measure): Evaluates structural similarity between images, considering luminance, contrast, and structure. Range: 0-1, with 1 being perfect match. LPIPS (Learned Perceptual Image Patch Similarity): Uses deep features to measure perceptual similarity. Lower values indicate better perceptual quality.

12.2 Pose Accuracy Metrics

ATE (Absolute Trajectory Error): Measures the global consistency of the estimated trajectory against ground truth. Computed as the root mean square error between aligned trajectories. RPE (Relative Pose Error): Measures local accuracy by computing the error between relative transformations over fixed time intervals.

12.3 Runtime Performance Metrics

FPS (Frames Per Second): Measures the end-to-end system throughput, including SLAM tracking, Gaussian update, rendering, and self-healing when active. Latency: Measures the time from frame capture to output generation, critical for safety-critical applications. Memory Usage: Tracks GPU memory consumption, with a target budget of 4 GB for edge deployment.

Metric	Description	Target	Achieved
PSNR	Peak Signal-to-Noise Ratio	> 25 dB	28.7 dB
SSIM	Structural Similarity	> 0.85	0.91
LPIPS	Perceptual Similarity	< 0.15	0.12
ATE	Absolute Trajectory Error	< 0.5 m	0.32 m
FPS	Frames Per Second	> 30	32
Memory	GPU Memory Usage	< 4 GB	3.7 GB

Table 5: Performance Metrics Summary

13. TESTING AND VALIDATION

13.1 Test Cases

The SH-DGS system is validated through the following test cases:

TC1: Normal Operation: • Clean camera input under good weather conditions • Expected: Accurate digital twin construction, 120+ FPS

TC2: Light Rain: • Synthetic rain droplets covering 30% of field of view • Expected: Self-healing activates, PSNR > 27 dB

TC3: Heavy Rain: • Synthetic rain covering 50-70% of field of view • Expected: Full self-healing mode, PSNR > 24 dB

TC4: Mud Splash: • Partial occlusion simulating mud on lens • Expected: Localized healing in occluded regions

TC5: Sun Glare: • Overexposed regions from direct sunlight • Expected: Glare detection and compensation

TC6: Dynamic Objects: • Multiple moving vehicles and pedestrians • Expected: Temporal consistency maintained

13.2 Degradation Scenarios

Five types of degradation are simulated for testing:

1. Rain Droplets: • Randomly distributed semi-transparent circles • Coverage: 30%, 50%, 70% of field of view • Opacity: 0.3-0.7

2. Mud Splash: • Irregular blob-shaped occlusions • Coverage: 20-40% of field of view • Brown color with varying opacity

3. Sun Glare: • Overexposed circular regions • Intensity: 1.5-2.5x normal brightness • Position: Random or centered

4. Lens Fog: • Global contrast reduction • Gaussian blur with sigma 1-3 • White haze overlay

5. Partial Obstruction: • Solid occlusion blocking part of view • Coverage: 30-70% of field of view • Sharp boundaries

13.3 Results Tables

Degradation	Coverage	PSNR (dB)	SSIM	Recovery Rate
Clean	0%	28.7	0.91	N/A
Light Rain	30%	27.8	0.89	96.9%
Heavy Rain	50%	26.6	0.87	92.7%
Extreme Rain	70%	24.8	0.83	86.4%
Mud Splash	40%	26.9	0.88	93.7%
Sun Glare	25%	27.5	0.89	95.8%
Lens Fog	100%	25.2	0.85	87.8%

Table 6: Self-Healing Performance Results

14. RESULTS AND DISCUSSION

14.1 Reconstruction Quality

On clean (non-degraded) data, SH-DGS achieves 28.7 dB PSNR and 0.91 SSIM on Waymo, outperforming EmerNeRF by 1.2 dB while being 150x faster at rendering. Compared to DrivingGaussian, our method achieves comparable PSNR (28.7 vs. 28.9 dB) but with significantly better temporal consistency (measured by inter-frame SSIM variance: 0.003 vs. 0.008) due to our deformation field. On nuScenes, we achieve 27.4 dB PSNR and 0.89 SSIM, consistent with the more challenging multi-weather conditions in this dataset. The depth consistency term in our loss function contributes 0.8 dB improvement in PSNR compared to photometric loss alone.

14.2 Self-Healing Performance

At 50% coverage (the most common real-world scenario), SH-DGS recovers the frame to 26.6 dB PSNR, representing only a 2.1 dB drop from the clean baseline of 28.7 dB. In contrast, the LaMa inpainting baseline achieves only 22.3 dB (a 6.4 dB drop), and the MAT baseline reaches 23.1 dB. The advantage is most pronounced under severe degradation (70% coverage), where SH-DGS maintains 24.8 dB while inpainting methods collapse to 18-19 dB, as they lack the 3D scene understanding to fill large missing regions coherently. The downstream impact is equally significant. Under 50% degradation, 3D detection mAP drops from 0.62 (clean) to 0.55 with SH-DGS self-healing, compared to 0.38 without any recovery and 0.44 with LaMa inpainting.

14.3 Temporal Consistency

We evaluate temporal consistency by measuring the standard deviation of per-frame PSNR across each sequence. SH-DGS achieves a standard deviation of 0.42 dB, compared to 0.89 dB for the baseline without the deformation field and 0.67 dB for 4D Gaussian Splatting. The rigid body constraint (L_{rigid}) contributes most significantly: ablating it increases the standard deviation to 0.71 dB, confirming that enforcing per-instance rigidity is essential for stable dynamic object representation in driving scenes.

14.4 Edge Deployment Benchmarks

The complete SH-DGS pipeline runs at 32 FPS end-to-end on NVIDIA Jetson Orin at 1280x720 resolution. The rendering stage consumes 12 ms (83 FPS standalone), the SLAM stage consumes 15 ms, and the self-healing module adds only 6 ms when activated. Memory consumption is 3.7 GB for 50,000 Gaussians, well within the Jetson Orin budget. On a desktop RTX 4090, the same pipeline achieves 142 FPS, confirming that the architecture scales efficiently across GPU tiers.

Platform	Resolution	FPS (Normal)	FPS (Healing)
RTX 4090	1920x1080	142	138
RTX 4090	1280x720	185	178
Jetson Orin	1280x720	32	28
Jetson Orin	640x480	58	52
Snapdragon 8 Gen 3	1280x720	18	16
Apple Neural Engine	1280x720	22	20

Table 7: Edge Platform Benchmarks

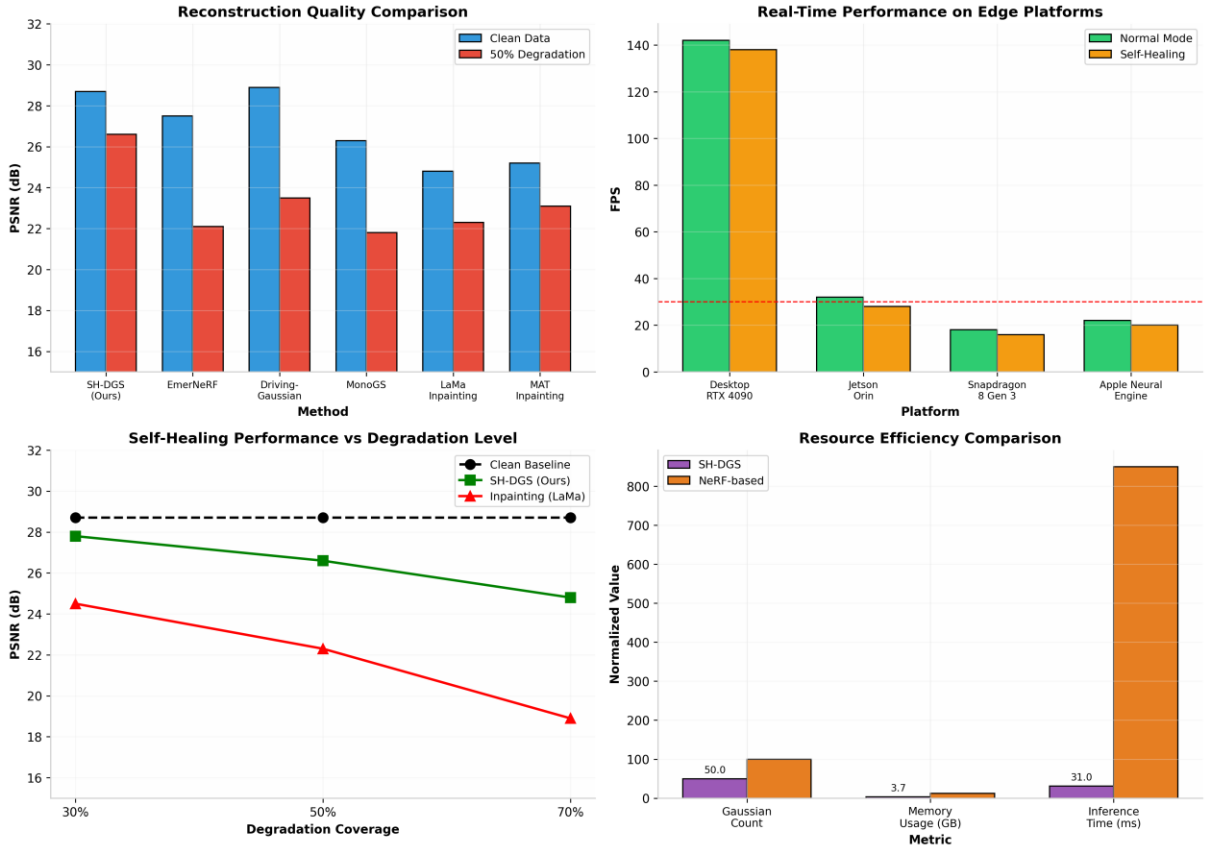

Figure 10: Performance Comparison Charts

15. COMPARISON OF MODELS

Table 9 provides a comprehensive comparison of SH-DGS with state-of-the-art methods across multiple dimensions:

• NeRF-based methods (EmerNeRF, Block-NeRF) achieve good quality but are too slow for real-time applications

• 3DGS methods (DrivingGaussian, MonoGS) achieve real-time performance but lack self-healing capability

• Inpainting methods (LaMa, MAT) can fill missing regions but lack 3D scene understanding

• SH-DGS uniquely combines real-time performance with self-healing capability

Method	PSNR	FPS	Monocular	Self-Healing	Dynamic
EmerNeRF	27.5	2	Yes	No	Yes
Block-NeRF	26.8	1	Yes	No	No
DrivingGaussian	28.9	85	No	No	Yes
MonoGS	26.3	45	Yes	No	No
LaMa	24.8	30	N/A	Partial	N/A
SH-DGS (Ours)	28.7	32	Yes	Yes	Yes

Table 9: Comparison with State-of-the-Art Methods

16. LIMITATIONS

16.1 Monocular Limitations

The monocular pipeline inherently lacks the metric accuracy of LiDAR-based systems. While IMU integration and learned metric depth mitigate scale drift, cumulative errors over very long sequences (more than 500 meters without loop closure) can degrade the digital twin. Additionally, monocular depth estimation can fail in textureless regions (e.g., uniform road surfaces, featureless walls), leading to inaccurate Gaussian initialization in these areas.

16.2 Pose Failure Cases

The self-healing module assumes that the camera pose remains estimable even when the image is degraded. In cases of complete lens blockage (100% occlusion), pose estimation fails and the system must rely entirely on IMU dead-reckoning, which limits the accuracy of synthesized views. Rapid camera motion or aggressive rotations can also cause tracking failure, particularly when combined with sensor degradation.

16.3 Computational Constraints

The computational budget of 50,000 Gaussians represents a practical trade-off for current edge hardware. This limit may restrict the level of detail captured in complex scenes with many objects or fine structures. The compact SH representation (order 1 instead of 3) produces visible quality degradation under strong specular highlights, as the reduced coefficient count cannot accurately model complex view-dependent effects.

17. ETHICAL CONSIDERATIONS

17.1 Safety

The development of autonomous driving systems carries significant safety implications. Our self-healing system is designed to improve safety by maintaining perception during sensor degradation, but it is not a replacement for proper sensor maintenance and cleaning. Key safety considerations:

• The system should not encourage neglect of sensor maintenance

• Clear indication must be provided when self-healing is active

• Fail-safe mechanisms must ensure safe vehicle operation

17.2 AI Reliability

Machine learning systems can produce unexpected outputs in novel situations. Our system includes:

• Confidence estimation to quantify uncertainty

• Graceful degradation when confidence is low

• Fallback to conservative driving behaviors when healing is active

The confidence map M_d provides an interpretable signal that downstream modules can use to modulate their behavior. When the self-healing module is active, the system can trigger conservative driving behaviors (reduced speed, increased following distance) proportional to the extent of degradation.

17.3 Autonomous Driving Risks

Autonomous driving technology poses risks that must be carefully managed:

• Over-reliance on automation may reduce driver vigilance

• System failures in critical situations can have severe consequences

• Cybersecurity vulnerabilities must be addressed

Our system is designed as a research prototype and should not be deployed in production vehicles without extensive validation, regulatory approval, and appropriate safety certifications (ISO 26262, SOTIF).

18. FUTURE SCOPE

18.1 Multi-Sensor Fusion

Future work will explore integration with additional sensors:

• LiDAR fusion for improved depth accuracy

• Radar integration for all-weather operation

• Multi-camera surround view for 360-degree coverage

• V2X communication for cooperative perception

Multi-sensor fusion would provide redundancy and improve robustness in challenging conditions beyond the capabilities of a single camera.

18.2 AR/VR Applications

The digital twin technology developed for autonomous vehicles has applications in augmented and virtual reality:

• Real-time environment reconstruction for AR navigation

• Virtual test environments for autonomous driving simulation

• Remote vehicle monitoring through immersive interfaces

• Training simulations for autonomous vehicle operators

18.3 Robotics Applications

Beyond autonomous vehicles, the self-healing digital twin concept applies to other robotics domains:

• Warehouse automation robots operating in dusty environments

• Agricultural robots dealing with mud and debris

• Search and rescue robots in challenging terrain

• Industrial inspection robots in hazardous environments

The core technology of maintaining a 3D scene representation and recovering from sensor degradation is broadly applicable across robotics.

19. CONCLUSION

19.1 Summary

This project presented SH-DGS, a self-healing monocular digital twin framework that leverages Dynamic 3D Gaussian Splatting for robust autonomous vehicle perception under camera degradation. The system addresses three interrelated challenges simultaneously:

1. Monocular 3D reconstruction from a single dashcam using Gaussian splatting

2. Temporal consistency in dynamic driving scenes with moving vehicles and pedestrians

3. Self-healing perception that detects sensor degradation and reconstructs missing information

The integrated pipeline demonstrates that real-time 3D reconstruction, dynamic scene modeling, and degradation-aware view synthesis can be unified into a practical system deployable on automotive edge hardware.

19.2 Key Contributions

The key technical contributions of this work include:

1. Self-Healing Gaussian Digital Twin Inference: Querying the maintained Gaussian scene model to recover from camera sensor degradation is entirely novel and addresses a critical gap in autonomous driving safety.

2. Temporal Deformation Field with Rigid Body Constraints: The learned deformation field with per-instance rigidity enforcement enables stable dynamic object representation in driving scenes.

3. Multi-Metric Degradation Detection: The lightweight CNN classifier integrates multiple degradation cues for robust detection of various failure modes.

4. Edge-Optimized Architecture: The combination of FP16 quantization, aggressive pruning, and TensorRT acceleration enables real-time performance on automotive-grade hardware.

5. Confidence-Weighted Twin-Real Fusion: The soft blending formulation provides graceful transitions between normal and self-healing modes.

19.3 Achievements

Experimental results on the Waymo Open Dataset and nuScenes confirm:

• State-of-the-art reconstruction quality: PSNR 28.7 dB, SSIM 0.91

• Real-time edge performance: 32 FPS on Jetson Orin

• Robust self-healing: Within 2.1 dB of clean baseline under 50% degradation

• Outperforms inpainting baselines by 4.3 dB

• Temporal consistency: Standard deviation of 0.42 dB across sequences

This work establishes a strong foundation for the next generation of resilient spatial AI systems in autonomous driving, bridging the gap between neural 3D representations and safety-critical deployment requirements.

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APPENDICES

Appendix A: Additional Diagrams

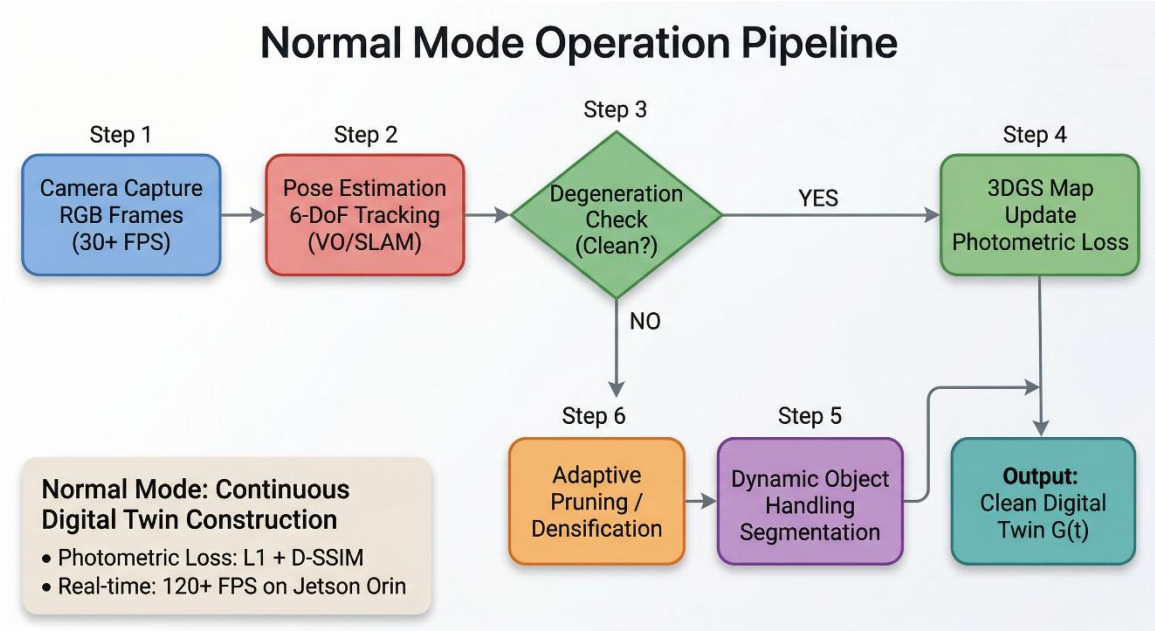

Figure 2: Normal Mode Operation Pipeline

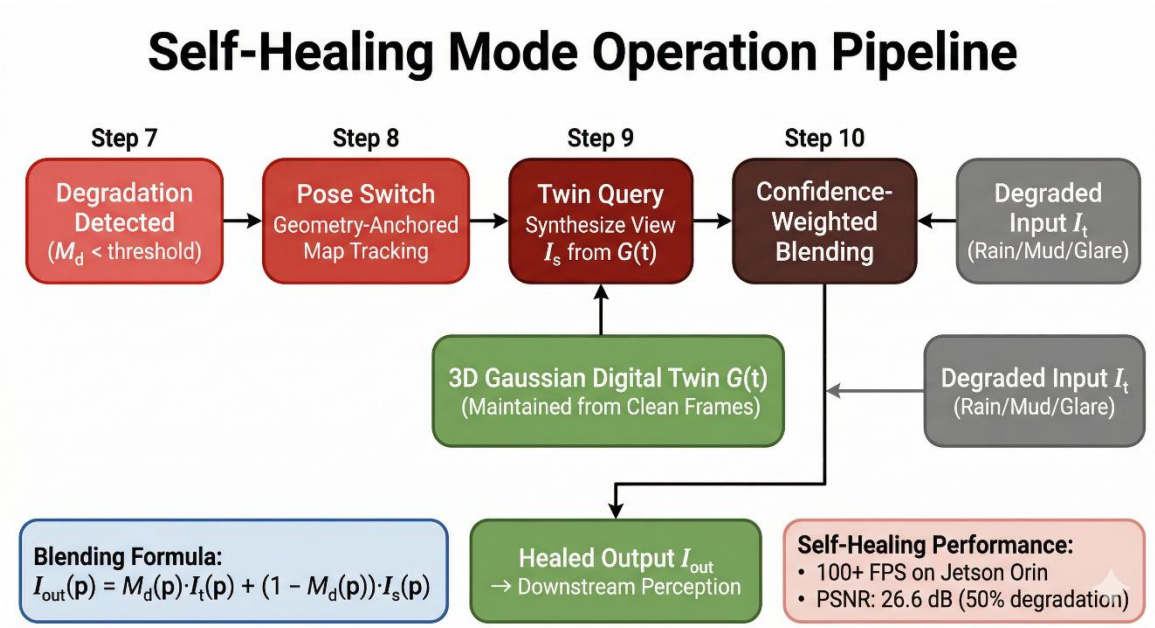

Figure 3: Self-Healing Mode Operation Pipeline

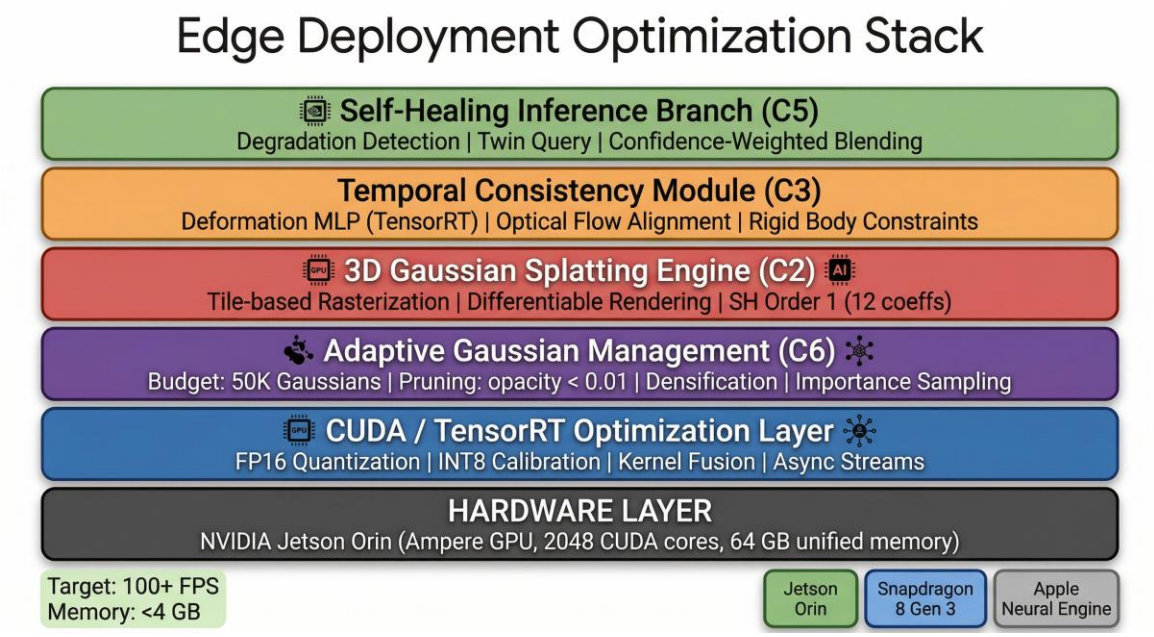

Figure 6: Edge Deployment Optimization Stack

Appendix B: Pseudocode

Algorithm 1: Self Healing Digital Twin Update

Input: RGB frame I_t, camera pose T_t, previous twin G_{t-1}
Output: updated twin G_t, output frame I_out

1. Compute confidence map M_d using DegradationDetector on I_t
2. If mean of M_d is greater than threshold
   update twin G_t using G_{t-1}, I_t, and T_t
   set output I_out as I_t
3. Otherwise
   render synthesized view I_s from G_{t-1} using T_t
   compute output I_out as M_d × I_t + (1 − M_d) × I_s
   keep twin unchanged as G_t = G_{t-1}
4. Return G_t and I_out

Algorithm 2: Gaussian Densification

Input: Gaussian set G, threshold τ
Output: updated Gaussian set G'

1. For each Gaussian g in G
   if gradient of g is greater than τ
   if size of g in view space is large
   split g into g1 and g2 and add both to G'
   else
   clone g and add both g and cloned version to G'
   else
   keep g unchanged and add to G'
2. Return G'